Ángel Alberto Carretero

angelalbertoc.r@gmail.com, letfunny.com

Experience

09/2025–present **Performance and Correctness – Office of the CTO**, Canonical (Remote)

My current focus is on correctness of distributed systems using formal (TLA+) and semi-formal methods (state machines). Some highlights:

- Implemented Raft in Rust in a very short timeframe through careful use of TLA+ for model-based testing of the code and discrete event simulation; both of which provided us with explicit guarantees about its correctness.
- Led some company-wide sessions on TLA+ with a focus on producing careful designs which substantially decrease the cost of implementing distributed systems.
- Used state machines in order to design and implement protocols such as snapshot installation for Raft. State machines allowed us to think upfront about correctness and produced code that was significantly easier to read and easier to debug.
- Designed a performant storage layer based on SQLite which uses a custom VFS.

Main skills: Rust and Tokio, systems programming, SQLite, TLA+, state machine-based design.

09/2023–present **Software Engineer – Office of the CTO**, Canonical (Remote)

Worked directly with the CTO to kickstart challenging projects used across Canonical. Some highlights:

- Lead developer of chisel, a tool written in Go to create containers with small storage footprint and a reduced attack surface.
- Contributor to dqlite, a distributed database using Raft and SQLite, written in C. I developed a snapshot protocol to bring followers up to date, and I used state machines to reason about termination.
- As part of my duties, I supervised and mentored developers, led planning, estimating, and architecting. In a few months I managed to deliver 1.0 for a project that had been stagnant for a year.

Main skills: Deep experience in Go and its runtime, containers and OCI images, GNU/Linux.

09/2022–09/2023 **Software Developer Engineer I – Books Detail Page**, Amazon (Madrid)

My team owned the book detail pages in all surfaces across www.amazon.com, a Tier-1 service that receives thousands of requests per second. Some highlights:

- Developed features using Java/Spring for the backend, JSPs and JS for the frontend. This required complex debugging across many different architectures and surfaces on a custom stack with dozens of layers and many different teams working on the same codebase simultaneously.
- Led a successful project coordinating two different teams in different organizations, PMs and design. Among my different duties, I carefully designed the experiment to ensure precise measurement of KPIs.
- Actively troubleshot issues, participated in maintenance and operational excellence including on-call duty for the main Amazon webpage.

Main skills: Full stack, driving features from experiment planning to production deployment.

09/2021 – 06/2022	**Researcher (Collaboration grant) – Computer Science department**, UAM (Madrid) Designed an esoteric programming language and created an interpreter from scratch in Rust which included a library of parser combinators, an interpreter, a typechecker, and more. **Main skills**: Rust, programming language theory.
06/2021 – 09/2021	**Software Developer Engineer Intern – Music ML**, Amazon (Berlin) Worked on recommendation systems for Amazon Music which used an orchestrator written in functional async Java. **Main skills**: Async functional Java, experiment design.

Education

2017 – 2022	**BSc in Mathematics**, Universidad Autónoma de Madrid
2017 – 2022	**BSc in Computer Science**, Universidad Autónoma de Madrid • Combined average of 9.02/10, 14 courses with honors (top 2%). Awarded three times the scholarship for the top 1% of all university students in Madrid. • Math Bachelor's thesis: *Pseudo-Riemannian Geometry and Schwarzschild spacetime*. (9.9/10) • Computer Science Bachelor's thesis: *Formal design and implementation of a programming language based on facets*. (9.5/10)

Talks

Git gud: how Git works internally – Open South Code 2024
Explore how Git stores its state on disk inside the .git subdirectory. For each command, inspect the results live using low-level tools to gain a deeper intuition.

Languages

Spanish	First Language.
English	C1 Level, certified Cambridge English Advanced.

Contact

Mail	angelalbertoc.r@gmail.com	LinkedIn	angel-alberto-carretero
Github	https://github.com/letFunny	Web	https://letfunny.com